CS4680 Prompt Engineering
AI Agent Project

Due Date
See the date in Canvas

Score
50

Questions and Directions

In this project, you will design and implement an AI Agent that receives output from a Large Language Model (LLM) and performs concrete actions based on that output. The agent should be able to interpret LLM responses and execute various tasks across different domains and applications. For reference, please see: https://youtu.be/wcS2QUXKeP4

Project Requirements

Your AI agent must include:

1. LLM Integration Module
    - Connect to an LLM API (OpenAI, Anthropic, Google, or others)
    - Send prompts and receive responses
    - Handle API errors and rate limiting
2. Action Interpreter/Executor
    - Parse LLM output to extract actionable commands
    - Convert LLM responses into executable operations
    - Execute actions across different domains (e.g., file operations, system commands, API calls, database operations)
3. User Interface
    - Terminal-based or GUI interface
    - Allow users to input natural language requests
    - Display results and execution status
    - Provide feedback on actions taken
4. Error Handling & Safety
    - Validate LLM outputs before execution
    - Handle errors gracefully
    - Implement safety checks (e.g., confirmation for destructive operations)
    - Log all actions for auditability

Submission Guidelines:

- **Code**: Submit your code repository link (GitHub URL).
- **Deployed URL**: Submit the URL to your application that we can test and run
- **Demo Video**: Submit a 1-2 min short demo video link (e.g., YouTube URL).

Grading Criteria:

- Problem Identification and Relevance (10%)
- Application Design and Implementation (30%)
- Use of Prompt Engineering Techniques (20%)
- UI/UX Design (10%)
- Presentation and Live Demo (30%)

Demonstration and Submission

Submission Requirements: each student must submit their project through a Google Form (https://forms.gle/nMHp5owwYidszwt58).